

Enterprise network design evaluation

Amari Bullard

UAT

NTW275

Prof. Sarah Bunce

Enterprise network design evaluation

I decided to create a new network for this assignment. I wanted to go with a large enterprise-level network. Large companies need reliable, and efficient networks to support their day-to-day operations.

The goal is to have the following

- High-speed connectivity for devices and applications
- Advanced network security for data protection
- The network needs to be scalable
- The network needs to be reliable with nearly no downtime

The components it has are

- Core network
 - Cisco Nexus 9000 switches - great for high performance and scalability for enterprise environments.
 - Cisco 3100 series firewall - designed for advanced threat protection and secure access
 - Dual ISP connections with load balancing for redundancy and traffic management
- Branch Offices
 - Cisco Catalyst 8000 edge platforms for their flexibility and low operational cost
 - Redundant links to maintain connectivity during an outage
- Wireless
 - Cisco Catalyst 9100 Wifi 6E APs for their AI integration and self-configuration for larger deployments
- Servers

- Cisco's USC X-Series modular system - optimized for AI workload and high performance and scalability
- Cisco Identity Services Engine for its robust authentication and policy enforcement
- Monitoring
 - Cisco DNA center for real-time network visibility and proactive issue-resolving
 - .Cisco Secure Network Analytics for threat detection and fast response

There are always positives and negatives to a network. This network specifically has a few different ones. The main performance strength is its high-speed links with the Nexus 9000 switches. The other is the quality of service for prioritizing important traffic. The main performance weakness is the latency. It can exceed 100 ms during peak hours. The security strengths are the end-to-end encryption with the IPsec tunnels and the MFA and VPN for remote access needs. The security weaknesses are the lack of a zero-trust architecture. It also lacks SIEM for more proactive threat protection. The scalability strength is the modularity which makes it easy to expand the network. The main weakness is the VLAN configs could reach capacity as the enterprise grows but this is unlikely and is easily solvable. The reliability strengths are the ISP redundancy and backup power for the main devices in the network. The weakness is the current lack of geographic redundancy in case of disaster.

Some of the improvements I would make personally are

- Performance

- 100 Gbps WAN links using dense wavelength division multiplexing
- CDNs for faster data delivery
- Security
 - Zero trust network access
 - Cisco SecureX for real-time threat monitoring and management
- Scalability
 - IPv6 for addressing
 - Possibly upgrading to 9600 Catalyst switches for expanded VLAN support but this is very expensive
- Reliability and redundancy
 - Implement cloud computing for redundancy using AWS or Azure
 - Have another data center in a separate location for geo-redundancy

Cost estimates

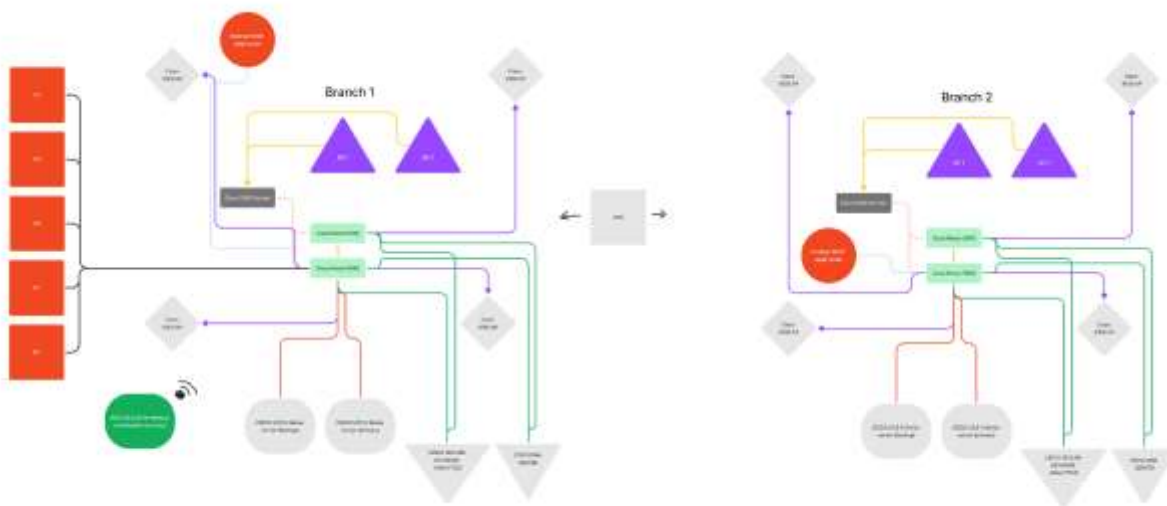
- Cisco Nexus \$8000 - \$15000 roughly
- Cisco Secure Firewall 3100 series - \$15,000- \$50,000 per unit for hardware and licenses
- Cisco Catalyst 9100 Access Points - \$700 - \$2500 per AP for hardware and the licenses
- Cisco USC X-Series Modular system - \$30,000 to \$100,000+ per unit depending on the configuration
- Cisco Identity Services Engine - \$20,000 to \$100,000 + annually depending on deployment size
- Cisco DNA Center - \$50,000 to \$150,000 + annually depending on deployment size
- Cisco Secure Network Analytics - \$100,000 to \$500,000 + annually depending on deployment size and features needed

- Dual ISP connectivity - \$500 to \$5000 + a month depending on the service level and bandwidth needed
- Secondary enterprise-level data center \$1 million to \$50 million + depending on size and specifications needed
- Disaster recovery services - \$10,000 to \$1 million + annually depending on the scale and complexity of the enterprise network

Implementation would be done in phases. Start with performance then move to security then scalability and then reliability.

There are some challenges. This network setup is extremely expensive. Using phased deployment is a good way to spread that cost out over different financial periods. Disruption is another issue. You can't disrupt your network during active hours. The best thing to do is schedule downtime during the least busy time on the network. Using the weekends is another good thing to do. Training staff to use new equipment can be hard especially if you are moving from one manufacturer to another. Having workshops and very in-depth documentation is the best way to avoid any major issues. Another challenge is making sure the equipment you buy is compatible with your existing equipment. You wouldn't want to waste a bunch of money on something that won't work. You need to look at documentation and do controlled testing before fully deploying anything new. Latency is another problem. A good way to avoid that is caching servers and optimizing QoS settings to help avoid any major delay issues.

In conclusion, the network and the improvements will fix any major weaknesses. Having a modern and efficient network is so important for a major enterprise-level business. The upgrades will help the business stay ahead instead of falling behind.



References

XAI. (n.d.). <https://x.ai/> (Cost analysis)

Cisco Nexus 9000 Series switches explained. (2024, October 15). [Video]. Cisco.

<https://www.cisco.com/site/us/en/products/networking/cloud-networking-switches/nexus-9000-switches/index.html>

Data sheets. (2024, October 31). Cisco.

<https://www.cisco.com/c/en/us/products/switches/nexus-9000-series-switches/datasheet-listing.html>

Cisco Next Generation Firewall (NGFW) demo. (2024, July 18). [Video]. Cisco.

<https://www.cisco.com/site/us/en/products/security/firewalls/secure-firewall-3100-series/index.html>

Cisco Catalyst 9100 Wireless Access Points - Cisco. (2024, November 20). Cisco.

<https://www.cisco.com/site/us/en/products/networking/wireless/access-points/catalyst-9100-series/index.html>

Cisco UCS X-Series X-Fabric Technology Sizzle video. (2024, November 26). [Video].

Cisco. <https://www.cisco.com/site/us/en/products/computing/servers-unified-computing-systems/ucs-x-series-modular-systems/index.html>

Cisco ISE licensing Guide. (2024, January 31). Cisco.

<https://www.cisco.com/c/en/us/products/collateral/security/identity-services-engine/ise-licensing-guide-og.html>

Cisco Catalyst Center overview demo. (2024, December 11). [Video]. Cisco.

<https://www.cisco.com/site/us/en/products/networking/catalyst-center/index.html> (DNA)

Network threat detection and response with Cisco Stealthwatch. (2024, December 10).

[Video]. Cisco. <https://www.cisco.com/site/us/en/products/security/security-analytics/secure-network-analytics/index.html>

Zola, A., & Burke, J. (2021, August 12). *dense wavelength-division multiplexing (DWDM)*. Search Networking.

<https://www.techtarget.com/searchnetworking/definition/dense-wavelength-division-multiplexing-DWDM>

What is a CDN (Content Delivery network)? | How do CDNs work? | Akamai. (n.d.).

Akamai. <https://www.akamai.com/glossary/what-is-a-cdn>

How To Implement Zero Trust: 5-steps Approach & its challenges | Fortinet. (n.d.).

Fortinet. <https://www.fortinet.com/resources/cyberglossary/how-to-implement-zero-trust>

Upgrading to IPv6 for scalability - Google Search. (n.d.).

https://www.google.com/search?q=Upgrading+to+IPv6+for+scalability&oq=Upgrading+to+IPv6+for+scalability&gs_lcrp=EgZjaHJvbWUyBggAEEUYOdIBCTM5MDE0ajBqNKgCALACAA&sourceid=chrome&ie=UTF-8

Poole, J., & Poole, J. (2023, December 6). Geo-Redundancy: a top business priority in the digital era. Interconnections - the Equinix Blog.

<https://blog.equinix.com/blog/2020/08/24/geo-redundancy-a-top-business-priority-in-the-digital-era/>

Network latency: common causes & proven tactics to reduce it | DigitalOcean. (n.d.).

<https://www.digitalocean.com/resources/articles/network-latency>

What is QoS? How does QoS help optimize network performance? (n.d.).

<https://www.site24x7.com/blog/optimize-qos>

